

TOLEDO SUBURBAN, MI, USA

WMO: 720275

Lat: 41.736N

Lon: 83.655W

Elev: 204

StdP: 98.90

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 04872

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.2	-13.7	-22.0	0.5	-16.6	-18.0	0.8	-12.5	9.5	0.9	8.5	-1.7	2.0	250	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	32.4	22.7	31.1	22.1	29.6	21.5	24.4	29.9	23.6	28.7	22.8	27.5	3.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	17.9	27.2	22.2	17.3	26.7	21.3	16.3	25.8	74.8	29.8	71.5	28.8	68.2	27.9	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.2	6.0	-20.2	34.4	3.3	1.6	-22.5	35.6	-24.4	36.5	-26.3	37.5	-28.7	38.6		
(5)			-20.6	25.8	3.1	0.8	-22.8	26.4	-24.7	26.8	-26.4	27.3	-28.7	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-3.0	-2.2	2.9	9.4	15.8	20.9	22.6	21.7	18.4	11.9	5.4	0.1		
	DBStd	10.36	6.32	6.05	5.82	4.89	4.90	3.55	3.22	2.95	4.14	4.89	4.98	5.27		
	HDD10.0	1551	403	344	233	68	6	0	0	0	1	34	153	308		
	HDD18.3	3330	660	576	478	271	111	14	3	6	51	208	388	564		
	CDD10.0	1696	1	2	14	50	186	327	392	362	251	93	16	2		
	CDD18.3	433	0	0	1	2	33	92	137	109	51	8	0	0		
	CDH23.3	3892	0	0	6	40	331	808	1278	926	430	71	2	0		
	CDH26.7	1196	0	0	1	4	94	264	434	256	127	16	0	0		
(14) Wind	WSAvg	2.5	3.2	3.0	2.9	3.2	2.4	2.1	1.7	1.5	1.6	2.3	2.7	3.0		
(15) Precipitation	PrecAvg	844	54	52	59	80	90	88	84	82	72	64	62	58		
	PrecMax	1129	98	124	127	157	161	147	178	210	135	163	173	136		
	PrecMin	661	15	13	14	23	32	27	31	18	28	4	22	12		
	PrecStd	120	23	29	28	30	43	33	35	42	34	35	31	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.9	17.3	23.7	27.3	31.6	33.8	34.9	32.6	32.7	29.0	22.1	16.0		
		MCWB	12.6	12.5	16.9	16.2	21.2	22.6	23.9	23.0	23.2	20.8	14.3	13.7		
	2%	DB	11.6	12.6	18.6	23.9	29.3	32.0	32.4	31.1	30.3	25.2	17.8	12.7		
		MCWB	9.9	10.2	13.2	14.9	19.6	22.0	23.0	22.5	22.0	18.0	12.9	10.9		
	5%	DB	8.6	8.6	15.3	21.4	27.3	29.9	31.0	29.5	27.7	22.6	16.0	9.9		
		MCWB	6.8	6.2	11.1	13.7	18.8	21.1	22.6	22.1	20.2	16.7	11.4	8.0		
	10%	DB	5.6	5.4	12.3	18.0	24.9	27.7	29.0	27.8	26.0	20.3	12.9	7.1		
		MCWB	3.8	3.1	9.0	11.9	17.8	20.0	21.7	21.1	19.7	15.7	9.4	5.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.8	13.9	17.3	18.0	22.5	24.4	25.6	25.3	24.3	22.3	16.5	14.2		
		MCDB	14.4	16.1	23.6	24.4	28.8	31.1	31.7	30.2	30.5	27.2	20.0	16.1		
	2%	WB	10.4	10.3	14.4	16.2	21.3	23.3	24.7	24.1	23.1	20.0	14.0	11.2		
		MCDB	11.3	12.5	17.5	21.6	27.3	29.2	30.3	28.6	28.2	23.9	17.2	12.5		
	5%	WB	6.9	6.1	11.8	14.6	20.0	22.4	23.8	23.2	22.0	18.0	11.9	7.9		
		MCDB	8.0	8.3	14.7	19.8	25.4	27.7	28.8	27.5	26.0	21.1	14.8	9.4		
	10%	WB	3.8	3.3	8.9	13.0	18.9	21.5	22.8	22.4	20.8	16.2	10.0	5.2		
		MCDB	5.5	5.3	12.4	18.2	23.8	26.1	27.4	26.4	24.1	19.8	12.9	6.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	7.8	8.9	10.7	11.1	10.7	10.8	10.9	11.4	10.5	8.5	6.4		
		MCDBR	10.0	11.5	13.7	15.3	13.7	12.6	11.9	12.2	13.4	13.6	12.3	9.7		
	5% WB	MCWBR	8.8	9.6	9.3	8.5	6.5	5.5	5.0	5.6	6.4	7.6	8.5	8.3		
		MCWBR	9.6	11.2	12.5	13.6	12.0	10.8	10.0	9.9	11.1	11.4	11.2	9.4		
(40) Clear-Sky Solar Irradiance	taub		0.297	0.334	0.360	0.405	0.432	0.451	0.443	0.433	0.385	0.360	0.322	0.311		
		taud	2.413	2.296	2.313	2.211	2.186	2.179	2.219	2.233	2.346	2.400	2.463	2.399		
	Ebn at Noon		853	864	883	864	847	829	831	828	847	827	817	799		
		Edh at Noon	79	104	115	137	144	146	139	133	110	93	75	74		
(44) All-Sky Solar Radiation	RadAvg		1.54	2.44	3.50	4.62	5.41	6.05	6.08	5.31	4.32	2.85	1.81	1.25		
	RadStd		0.16	0.32	0.32	0.40	0.47	0.40	0.33	0.20	0.35	0.24	0.24	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+87	N/A

Nomenclature: See separate page